Why Cold Energy Settles

A tethered medium whose displacement is the gravitational polarization

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The question. Galaxy rotation curves follow a single relation between observed and baryonic acceleration. In the framework studied here the acceleration scale is set by dark energy:

$$a_0 = \kappa c\sqrt{G\rho_{\rm DE}} \simeq 0.173\, cH_\infty, \qquad \kappa = \tfrac{1}{2}\ \text{(fitted)}.$$

Clusters, the Bullet Cluster, Local Group timing and the cosmic web also require a cold, collisionless mass component, "cold energy", with $\Omega_c/\Omega_b = 5.364$. Why should that component arrange itself, galaxy by galaxy, into exactly the profile the acceleration relation demands? This note gives the mechanism in its simplest form.

1. The medium and its displacement. Treat cold energy as a cold medium of real rest mass. Let $\mathbf{P}$ be its displacement field:

$$\partial_t \mathbf{P} = -\mathbf{j} \quad \Longrightarrow \quad \nabla\cdot\mathbf{P} = \rho - \bar{\rho}.$$

So the "polarization" of the gravitational vacuum is not an extra field: it is how far the medium has moved. This is a gravitational analogue of a dielectric, close in spirit to Blanchet's dipolar dark matter [2] and to superfluid dark matter [3].

2. The action. With $K = 4\pi G$ and $u = K|\mathbf{P}|$,

$$\mathcal{L} = -\frac{|\nabla\Phi - K\mathbf{P}|^2}{2K} - \rho_b\Phi - W_{\rm rem}(u) - \frac{\gamma}{2}|\nabla\mathbf{P}|^2, \qquad W_{\rm rem} = W_{\rm full} - \frac{u^2}{2K}, \qquad K\frac{dW_{\rm full}}{du} = \frac{a_0 u}{a_0 - u}.$$

The vacuum term saturates at the wall $u \to a_0$. (The subtraction in $W_{\rm rem}$ is required; pairing the completed square with $W_{\rm full}$ double-counts the quadratic term.) The stiffness $\gamma = KL_*^2/2$ is the one extra constant, $L_* \in [5.75\,\text{pc}, \gtrsim 1\,\text{kpc}]$. It is not optional: without it the Sun's own field carries the simple function's constant a_0 tail, which the planetary ephemerides exclude by $\sim 2600\times$. It is irrelevant on galaxy scales.

3. The tether. Varying Φ gives the Newtonian flux of the baryons, $g - u = g_N$. Varying $\mathbf{P}$ gives the balance of the medium. Its self-gravity cancels against the quadratic part of the field energy, so each element feels only

$$\underbrace{g_N}_{\text{baryons pull in}} \qquad \text{against} \qquad \underbrace{R(u) = K\frac{dW_{\rm rem}}{du} = \frac{u^2}{a_0 - u}}_{\text{vacuum pushes back}}.$$

Like a mass on a spring, the medium comes to rest where the two balance. The balance point is unique (the energy is convex) and gives

$$g_N = R(u) \implies u = \frac{a_0 g}{a_0 + g} \implies \boxed{g_N = \frac{g^2}{g + a_0}}$$

This is the "simple" interpolating function [1]: $g \to \sqrt{a_0 g_N}$ for $g_N \ll a_0$, giving $V_f^4 = GM_b a_0$; and $g \to g_N$ for $g_N \gg a_0$. We checked it numerically: $\nu = g/g_N = 100.5$ at $g_N = 10^{-4}a_0$. **Cold energy settles because the law's profile is the only state in which the medium is at rest.**

4. How it gets there. Galaxies assemble over $\gtrsim 10^9\,\text{yr}$, far slower than the medium's response time. The medium therefore follows the baryons adiabatically, and essentially no energy is radiated ($\leq 10^{-195}$ of the settling energy). A sudden start would instead relax by emitting longitudinal waves of $\mathbf{P}$, but that case is not realised in nature. The tether's projected capacity is $\bar{\Sigma} \leq a_0/\pi G$ ($\approx 214\,\text{M}_\odot\,\text{pc}^{-2}$). The law's own profile never reaches it.

5. What settles, and what does not.

- **Galaxies.** Nearly all baryons are condensed (stars and cold gas), and the medium tethers to them. This reproduces the radial acceleration relation: 0.100 dex on raw SPARC.
- **Solar System.** The stiff medium cannot bend on scales below $\sim 2\,\mathrm{pc}$, so the Sun is screened even inside the Galaxy's field. Cassini's quadrupole comes out at $\leq 3.5\times10^{-6}$ of the 2026 bound [4], even though the Galactic boost at the Sun is kept at 35–57%. The only residue is a uniform sunward pull $GM_\odot/L_*^2 \leq 4.2\times10^{-15}\,\mathrm{m\,s^{-2}}$. Wide binaries come out Newtonian (-0.8σ against Gaia DR3); cluster satellites keep their boost.
- **Large scales.** Two rules (each halo tethers only its own supply, and only condensed baryons source the tether) make the framework's own simulations consistent with raw KiDS-1000 cosmic shear: $\Delta\chi^2 = +2.03$ against a same-pipeline control.
- **Mergers.** At collision speeds the tether cannot follow, so cold energy is collisionless. This is consistent with the Bullet Cluster.
- **Hot-gas systems (prediction).** Groups and clusters are dominated by hot, pressure-supported gas. We predict that much of their cold energy stays *untethered*: collisionless and concentrated in the core. This is a candidate for the extra core mass seen in cluster lensing ($\times1.8$ inside $\sim1\,\mathrm{Mpc}$, 7σ). A frozen test is queued.

What is not yet derived.

- Why only condensed baryons tether. The action as written responds to all baryons.
- The energy-export channel and the frame anchor.
- The origin of the medium's mass, possibly set by $\rho_{\rm DE}$.
- The relativistic completion, needed so that lensing is predicted rather than assumed.
- κ is fitted, L_* is only bounded, Ω_c/Ω_b is measured, and massive early-type galaxies still show unexplained core mass.

Sealed predictions that can kill it. (i) A Cavendish experiment whose source acceleration is changed $\times4$ must see $|\Delta G/G| < 10^{-6}$; MOND-boosted lab models predict $\sim 3\times10^{-4}$. (ii) A probe beyond 1000 AU must see no non-thermal anomaly $\geq 10^{-13}\,\mathrm{m\,s^{-2}}$. (iii) A Sun–Earth saddle-point flyby sees nothing. (iv) Gaia DR4 wide binaries are Newtonian.

What is new here. The interpolating function [1] and the polarization picture [2,3] are not new. New are: the scale is tied to dark energy, so a_0 is flat in redshift if $\rho_{\rm DE}$ is constant (testable with ALMA, Nov 2026); the medium-is-the-polarization identification, which makes the law the tether's unique rest state; EFE-blindness from free-fall basins (Gaia DR4, Dec 2026); and the condensed-only, own-supply settling that passes raw cosmic shear.

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[1] B. Famaey & J. Binney, MNRAS 363, 603 (2005). [2] L. Blanchet, Class. Quantum Grav. 24, 3529 (2007); L. Blanchet & A. Le Tiec, Phys. Rev. D 78, 024031 (2008). [3] L. Berezhiani & J. Khoury, Phys. Rev. D 92, 103510 (2015). [4] R. S. Park et al., arXiv:2602.17884 (2026). Companion notes: PAPER47 (doi:10.5281/zenodo.23354043), PAPER48 (doi:10.5281/zenodo.23372002).
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